



Ultraviolet Light-Driven gaining of hydroxyl and nitrogen oxide radicals in Plasma-Treated water

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ABSTRACT

Here we numerically demonstrate that photolytic processes induced by external ultraviolet (UV) sources considerably affect the chemical properties of plasma–water systems. When describing the chemical transport between surface dielectric barrier discharge (SDBD) and deionized water, the plasma is simulated using a global model, while the gas (air) and liquid (water) regions are considered in a one-dimensional space. Our simulation involving exposure of plasma-treated water (PTW) to UV shows that long-lived species in PTW are readily photolyzed and form hydroxyl radical (OH) and nitric oxide (NO) over the entire liquid region. The total OH concentration was enhanced by 11.6 times at 100 s under exposure to a conventional UV lamp, while the total NO concentration was remarkably enhanced up to 45.8 times at 300 s. The time at which the OH concentration notably increases was advanced by more than 100 s by photolytic processes. The results directly reveal that the exposure of PTW to UV is an attractive approach to selectively releasing extra OH and NO, thereby increasing the PTW reactivity and applicability. Owing to their high reactivity and oxidizing power, the short lifetime of radicals including OH and NO in PTW, which is crucial in diverse applications, has been troublesome. In this regard, our findings will provide new guidance in utilizing PTW for various applications.

1. Introduction

Reactive oxygen and nitrogen species (RONS) are now invaluable in a wide range of scientific and industrial applications, from microscopic [1–3] to macroscopic perspectives [4,5]. Owing to the increasing need for decentralizing chemical production with nontraditional techniques, atmospheric-pressure low-temperature plasmas have been actively investigated as a pragmatic tool for on-demand RONS production. The most attractive property of such plasmas is that high-energy electrons in plasma facilitate “high-temperature” chemical processes at room temperature on thermally sensitive materials. Thus, using plasmas in the fields of biomedicine, food, and agriculture associated with RONS activity has been growing extensively.

In such applications, because plasma contact with water or at least is placed in water-containing wet environments, physical and chemical processes in plasma–water systems have become more important [6–8], leading to better utilization of the plasma systems. Thus, complex physicochemical processes that occur at the plasma–liquid interface have been actively studied, bridging plasma and plasma-induced phenomena inside liquids [8–15]. In the meantime, plasma-treated water

(PTW) has been treated as a fundamental research topic for model experiments or simulations.

Plasma-treated water or water-containing material is a cocktail of reactive chemical species, and their interplay with plasma species makes the PTW more complex and hinders its applications [10]. Selective production and activation of chemicals in PTW are nontrivial and crucial for target applications because each chemical species has a different lifetime, reactivity, oxidation power, and biochemical characteristic. Thus, earlier studies have been achieved by adjusting plasma and/or water properties [10,16,17] or using chemical scavengers [18]. Ultraviolet (UV) light-assisted photochemical engineering has been well-established in diverse practical applications as an advanced technique [19–22]. However, utilizing an external UV source to activate RONS in water with plasmas has been rarely reported.

Herein, this numerical study proposes a novel approach to control RONS in the plasma–liquid system and PTW by external UV irradiation. We found hydroxyl radical (OH) and nitrogen oxide (NO) are effectively produced by the photochemical reaction of chemicals in PTW with UV light [4,23], indicating that external UV irradiation largely enhances the functionality of PTW. In particular, the photochemical production of OH

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in PTW is focused on in this work. This diatomic radical, which is an important intermediate substance, is crucial in the PTW chemical kinetics because of its high reactivity and oxidizing power, making PTW applicable in diverse areas. Furthermore, OH is the most effective species in PTW to disinfect biological substances and bacterial inactivation [9,24,25].

Our chemical transport model simulates a plasma-water system, consisting of a surface dielectric barrier discharge (SDBD) in air and pure water. To make computational modeling of the plasma–water system more realistic, convective flow and heat transfer are considered along with the water evaporation in the system. With this model, we obtain the variation of RONS in PTW, especially OH, NO, and O₃, with and without UV photochemical processes.

2. Plasma chemistry model

Recently, computational studies have investigated RONS dynamics in plasma–liquid systems [11,26–28]. Chemical kinetics during the direct interaction between SDBDs and micrometer-scale water layers on tissue were reported [11]. The reactive species distribution in air and water with an indirect interaction between the air discharge and water was demonstrated [27,28]. In such studies, the size of plasma generated on a dielectric surface is much smaller than the entire system. Therefore, a zero-dimensional (0D) global model has been commonly employed to simulate SDBD and a 1D transport model has been applied to identify chemical transport from the gas to the liquid regions. These indirect plasma treatments can minimize unexpected physicochemical phenomena, such as turbulence in air and liquid, transient discharges, impurities emitted by an electrode, and damage to the intended target. However, there is still an imbalance between experiments and numerical modeling partly due to simple approximations of temperature, convection, and humidity in plasma–liquid systems, which make the modeling far from reality.

We implemented an advanced computational simulation based on a 1D transport model to better describe an indirect plasma interaction in the plasma–water system. Specific modeling efforts include the following: i) heat transfer to consider increasing gas temperature in the plasma region caused by ohmic heating; ii) convective transport of chemical species induced by the electrohydrodynamic (EHD) flow; iii) water evaporation owing to the increased temperature at the water surface. UV photolysis which is the crucial part of our modeling is listed in Table 1.

The numerical modeling was performed for three different regions: plasma (0D), gas (1D), and liquid (1D) regions. In Fig. 1, a simplified diagram of the plasma–water system described in the numerical simulation is shown with the numerical procedure.

2.1. Plasma region

Multiple time steps are required to avoid time-consuming and costly simulations because of the complex dynamics of the plasma species over various time scales from nanoseconds to minutes. This computational

Table 1

Photolysis and related reactions are considered in the liquid region of our model. All reaction rate coefficients were calculated based on the measured UV lamp spectrum. The relevant absorption coefficients and quantum yields used to calculate the rate coefficients are found in references.

No.	Reaction	Rate coefficient [s ⁻¹]	Reference
1	O ₃ + hν (+H ₂ O) → H ₂ O ₂ + O ₂	0.0784	[41,46]
2	NO ₂ + hν → NO + O [•]	1.98 × 10 ⁻³	[42]
3	NO ₃ + hν → NO ₂ + O [•]	2.88 × 10 ⁻⁴	[43]
4	NO ₃ + hν → ONOO ⁻	3.03 × 10 ⁻⁴	[43]
5	NO ₃ + hν → NO ₂ + O	1.05 × 10 ⁻⁴	[43]
6	H ₂ O ₂ + hν → OH + OH	0.0143	[44]
7	HONO + hν → NO + OH	0.0690	[45]

strategy enables us to numerically disentangle the slow and fast particle dynamics in plasma. First, similar to the previous global models [27], the governing equations for an SDBD plasma comprise mass and energy conservation equations for a single period, τ_{rep} , of 62.5 μs corresponding to a repetition frequency of 16 kHz, which was used in our experiment [29]:

$$\frac{\partial n_{p,g}}{\partial t} = \sum_j k_j \prod_r n_r \pm \frac{\Gamma_{p \leftrightarrow g}}{L_{p,g}} \quad (1)$$

$$\Gamma_{p \leftrightarrow g} = \frac{2D_g(n_p - n_g)}{L_p + L_g} \quad (2)$$

where $n_{p,g}$ is the number densities of the gas species in the plasma and gas (afterglow) regions, respectively; k_j is the rate coefficient of the j th reaction; n_r is the number density of species involved in the reaction related to $n_{p,g}$; $\Gamma_{p \leftrightarrow g}$ is the diffusion flux between the plasma and gas regions; and D_g is the gas diffusion coefficient, which depends on the gas temperature, T_g . The plus-minus signs on the right-hand side of Eq. (1) denote the direction of the diffusion flux. In this model, 53 chemical species and 624 relevant reactions were used for describing the plasma region, which are found in Table 1 of Ref. [27]. The thicknesses of the plasma (L_p) and gas (L_g) regions were set to 0.1 and 50 mm, respectively.

Instead of solving Poisson's equation as in a self-consistent model, a Gaussian-like pulsed electric field was defined, which is commonly assumed to zero dimensionally describe the streamer discharge [30] as follows:

$$|E| = E_0 \exp \left[-\frac{1}{2} \left(\frac{t - 5\tau_{\text{pls}}}{\tau_{\text{pls}}} \right)^2 \right] \quad (3)$$

where E_0 is the peak value of the electric field and $\tau_{\text{pls}} = 1$ ns is the pulse width, which defines the duration of the electric field induced by a single streamer at the dielectric surface [30]. The coefficients of the electron-related reactions and the electron mobility were obtained as functions of the electric field strength $|E|$ and water vapor concentration using BOLSIG+ [31]; thus, they were used as inputs to the 0D plasma model via two-dimensional linear interpolation. Based on the single-period result, the cycle-averaged power density (P_{cal}) is calculated as follows:

$$P_{\text{cal}} = \frac{L_p}{\tau_{\text{rep}}} \int_0^{\tau_{\text{rep}}} \sum_k Z_k \mu_k n_k |E|^2 dt \quad (4)$$

where Z_k and μ_k are the absolute charge value and mobility of the charged particle " k ," respectively. For maintaining a constant power density (P_{cal}) at $P_{\text{set}} = 200 \text{ mW}\cdot\text{cm}^{-2}$, the electric field amplitude (E_0) was optimized for the next cycle using $E_0 = \alpha(P_{\text{set}} - P_{\text{cal}})$ for every pulse period, where α is an optimization parameter. The iteration is repeated until the result reaches a periodic steady state. The calculated results hold under the conditions described in the result section. Nevertheless, the developed model can predict changes in the reaction kinetics according to the driving frequency and power delivered to the plasma.

2.2. Gas and liquid regions

The transport of ionic species from the plasma region toward the air and water is very limited by their short lifetime and diffusion length, which is typically less than 10 μs and 100 μm , respectively [27]. Therefore, charged species exist only in the vicinity of the plasma and the liquid surface charge is almost non-existent. After the calculation for the short period, τ_{rep} , 1D continuity equations were solved for the long-term (1 s) neutral species dynamics in the gas and liquid regions with 21 species and 63 reactions in the gas and 33 species and 109 reversible and irreversible reactions in the water [27,28]:

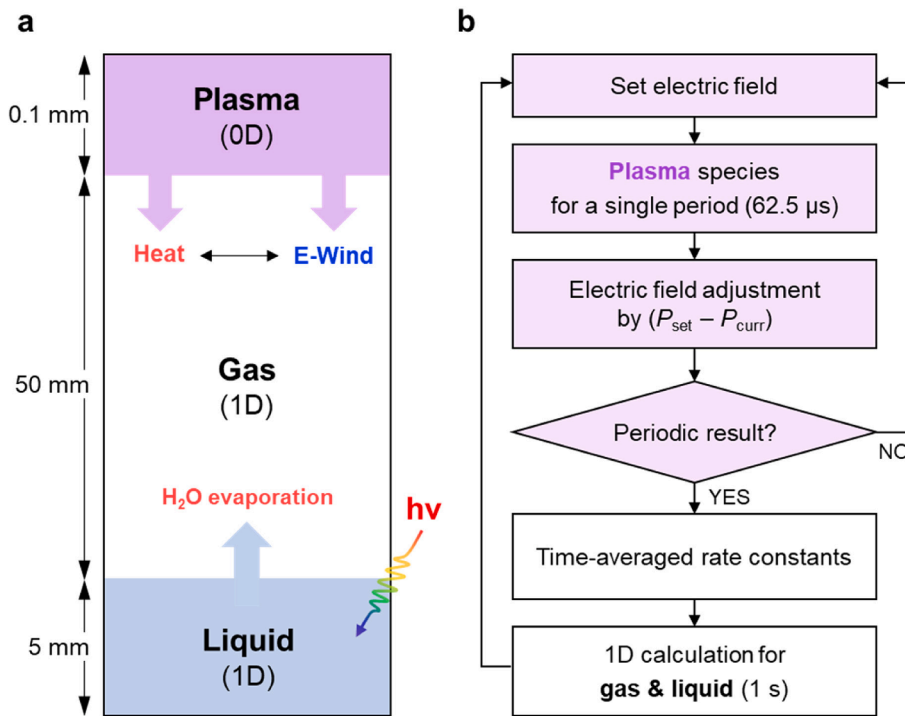


Fig. 1. (a) Schematic of the plasma–water modeling. A 0.1-mm-thick plasma layer was applied to calculate the diffusion flux (Γ_{pg}) of plasma species between the plasma and gas regions and the thicknesses of the air (L_g) and the liquid (L_l) regions were 50 and 5 mm, respectively. UV photolysis of the chemicals listed in Table 1 was considered in the liquid region. (b) A flow chart of the computational procedure presents a feedback loop for a 62.5- μ s-period discharge and subsequent steps in the gas and liquid for 1 s.

$$\frac{\partial n_j}{\partial t} = \sum_j k_j \prod_r n_{j,r} + D_j(T_{g,l}) \nabla^2 n_j - \mathbf{u} \cdot \nabla n_j \quad (5)$$

where n_j is the concentration of gaseous and aqueous species ‘j,’ $\mathbf{u}$ is the fluid velocity by convection, which was considered in the gas region, and $T_{g,l}$ are the gas and liquid temperatures, respectively. The diffusion coefficients are proportional to T_g^2 at low temperatures and $T_g^{1.65}$ at high temperatures [32]; thus, we approximate $D_j(T_g)$ as $D_j(300[K]) \times (T_g/300[K])^{1.82}$ [26]. As discussed, the convective flow is induced by natural convection due to the gradient of T_g , as described in Eq. (6). In addition, considering that the EHD flow is generally induced by surface plasma at a speed of a few m/s [33–35], an empirical EHD flow speed value of 0.5 m/s was applied to $\mathbf{u}$ in the model.

The governing equations for gaseous species also involve the heat transfer equation to consider the heat flux due to plasma’s ohmic heating coupled with the convective flow. Heat transfer in the gas region was solved in the form of

$$A_c \rho C_p \frac{\partial T_g}{\partial t} + A_c \rho C_p \mathbf{u} \cdot \nabla T_g = A_c k \nabla^2 T_g + A_p P_{set} + Q_{out} + Q_{vap} \quad (6)$$

where A_c is the cross-sectional area of the 1-D model, ρ is the density of the gas (air), C_p and k are the heat capacity and thermal conductivity of gas, respectively, and A_p is the cross-sectional area of the SDBD plasma. Additionally, Q_{out} is the heat loss at the wall boundaries contacting the reservoir air (at 300 K), and Q_{vap} is the heat flux by water evaporation. With configurations of the SDBD in air, most of the electric power is dissipated by ohmic heating in a narrow discharge layer, and the gases from the discharge layer and dielectric plate are directly heated by ohmic heating and power transfer [26]. Therefore, $A_p P_{set}$ is added as a heat source term in the equation. The heat loss term Q_{out} involves two heat flux components: convective heat transfer between ambient air and the dielectric plate and that between ambient air and the metal chamber sidewall. Additionally, heat transport by the evaporation of water at the gas–liquid interface is considered in the following way [36]:

$$Q_{vap} = -D_{H_2O,g} \left. \frac{\partial C_{H_2O,g}}{\partial x} \right|_{x=interface} \cdot H_{vap} \quad (7)$$

where H_{vap} is the latent heat of the evaporation of water. $C_{H_2O,g}$ at the gas–liquid interface is determined by its partial pressure from Antoine’s equation [37]:

$$p_{H_2O} = 8.07131 - \frac{1730.63}{233.426 + T_{inter}} \quad (8)$$

where p_{H_2O} is the partial pressure of H_2O , T_{inter} is the temperature at the gas–liquid interface, and the constant values were taken from ref. [38]. The water evaporation equations were coupled to the gaseous H_2O species governing equations and to the heat transfer equation, Eqs. (5) and (6), respectively.

Based on the two-film theory [39], which explains mass transfer at the interface between two stagnant film layers with different phases, the diffusion flux of gaseous species dissolving into the water $\Gamma_{g \rightarrow l}$ can be expressed as

$$\Gamma_{g \rightarrow l} = D_{adj} (H n_g - n_l) / \delta x_{liq} \quad (9)$$

$$D_{adj} = \frac{D_g D_l \delta x_{liq}}{D_g \delta x_{liq} + D_l \delta x_{gas} H} \quad (10)$$

where H is Henry’s law constant, and δx_{gas} and δx_{liq} are the thickness of the gas and liquid films, respectively. As δx_{gas} and δx_{liq} vary in the range of 1 – 10 mm and 10 – 100 μ m, respectively [40], $\delta x_{gas} = 10$ mm and $\delta x_{liq} = 100$ μ m were applied in this work. The adjusted diffusion coefficient D_{adj} is considered to impose flux conservation on the interface, which is more physical than $n_l = H n_g$ that indicates monotonically increasing n_l with time. Note that this theory is valid with a laminar gas flow and a stable interface, which is addressed in our model. The flux with D_{adj} calculates a time-varying $\Gamma_{g \rightarrow l}$ at the interface taking into account the difference in diffusivity of the two regions and the solubility of each species. We can easily find that the relation of $n_l = H n_g$ is usually employed in previous works, which means the flux is only proportional to the solubility of each species. However, $\Gamma_{g \rightarrow l}$ calculated with D_{adj} is related to both Henry’s law and flux conservation. Therefore, even in the case of a highly soluble chemical species, D_{adj} regulates dissolution into the water when the concentration reaches equilibrium at the gas–liquid

interface.

2.3. UV photolysis

As aforementioned, the modeling includes the UV photolysis reactions expected in the plasma–water system (Table 1). The UV photolysis rate Q of chemical species in the liquid region was obtained as follows:

$$Q = \int \Phi(\lambda, T_{\text{liq}}) \sigma(\lambda, T_{\text{liq}}) F(\lambda) d\lambda \quad (11)$$

where $\Phi(\lambda, T_{\text{liq}})$ is the quantum yield of the UV photolysis and $\sigma(\lambda, T_{\text{liq}})$ is the spectral absorption cross-section at the wavelength λ of light and liquid temperature T_{liq} . $\Phi(\lambda, T_{\text{liq}})$ and $\sigma(\lambda, T_{\text{liq}})$ are provided in Fig. 2a and Table 1 with the references. The measured irradiance per unit wavelength of a 6-W UV lamp (Spectroline ENF-260C/FE) was used as $F(\lambda)$. The lamp data was measured by using a combination of a spectrometer (Ocean Optics MAYA2000 Pro) and an optical fiber (Ocean Optics QP400-2-SR). As seen in Fig. 2b, the light system was equipped with two UV lamps, comprising UV-C and UV-A spectra. The UV lamp irradiated only in the liquid region, and it was assumed that UV was irradiated uniformly in the entire liquid region. The peak of the UV-C lamp spectrum mainly overlapped with the absorption profiles of $\text{O}_3(\text{aq})$ and hydrogen peroxide $\text{H}_2\text{O}_2(\text{aq})$, whereas that of the UV-A lamp was related to nitrite $\text{NO}_2^-(\text{aq})$ and nitrous acid $\text{HONO}(\text{aq})$. The ozone production by the UV lamps is negligible because the UV intensity below 220 nm of the lamps used in this work is almost zero [47].

The described numerical simulation was implemented using MATLAB [48] and COMSOL Multiphysics [49]. The initial conditions of air were 77.5% N_2 , 20.8% O_2 , and 1.7% H_2O (corresponding to 50% relative humidity) at 300 K. The H_2O concentration was continuously changed by increasing the gas temperature and consequent evaporation of water. In the liquid, the initial concentrations of H^+ and OH^- were $6.02 \times 10^{-19} \text{ m}^{-3}$ to obtain a pH of 7 at 300 K, whereas the concentrations of the dissolved N_2 and O_2 were obtained using Henry's law. The initial concentration was set at 10^5 m^{-3} for all species, except N_2 , O_2 , and H_2O in the plasma and gas regions and N_2 , O_2 , H_2O , H^+ , and OH^- in the liquid region.

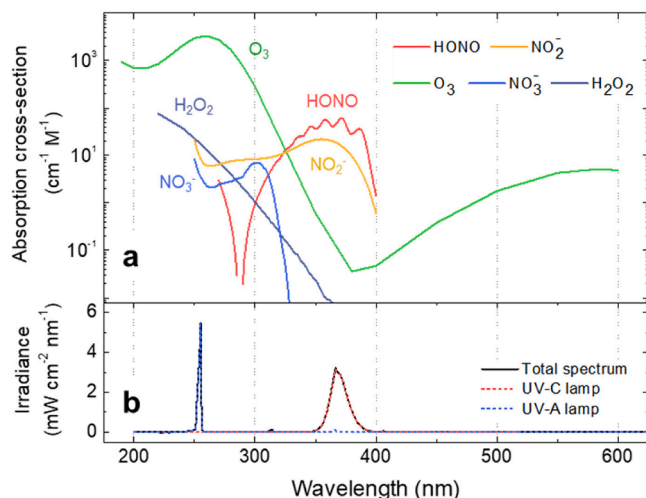


Fig. 2. (a) Absorption cross-section data of aqueous species in the spectral range of 200–600 nm [41–46]. The data were used for the UV photolysis reactions of each species. (b) Spectral distribution of irradiance with UV-C and UV-A lamps. The total spectrum using both lamps was used for the UV photolysis reactions.

3. Results and discussion

3.1. Validation of the modeling

To validate our advanced numerical model, the computed concentrations of gaseous O_3 and nitrogen trioxide (NO_3) were compared with the experimental result measured in our previous work; the details of the experimental setup, condition, and results can be found in Ref. [29]. The initial condition used in this work was identical to the experimental conditions of the previous work. Fig. 3a and 3b show the experimental and numerical results of the O_3 and NO_3 concentrations, respectively, at a distance of 50 mm from the SDBD. The time variations of calculated concentrations show a good agreement with the experimental data. However, both O_3 and NO_3 concentrations obtained from the numerical results exceed the experimental values by approximately-four times. This discrepancy seems reasonable because our 1D transport model could not account for the area difference between the two regions. The cross-sectional areas of the surface DBD and the gas regions were 49 and 225 cm^2 , respectively; therefore, the simulation results possibly exceed the experimental values because of diffusion along the horizontal direction according to the difference in width.

3.2. RONS concentration distribution

First, we considered the spatial distribution of the three important species, OH, O_3 , and NO, at a given time. Fig. 4 shows the spatial distributions of RONS in the presence of UV irradiation in the liquid region at the early stage (100 s) and long after plasma onset (300 s). The concentration of these species in the gas region is provided in Supplementary Fig. 1; the change in concentration in the gas region due to UV irradiation in the liquid region is negligible. The OH and NO concentrations are less than 10^{15} m^{-3} in the gas region. The concentration of O_3 is about $2.6 \times 10^{23} \text{ m}^{-3}$, the highest among the reactive species.

Fig. 4a compares the spatial profiles of the OH, O_3 , and NO concentrations in the liquid at 100 s after turning on the plasma. Here, two cases with (blue) and without (red) UV exposure to the liquid are plotted. Owing to the UV irradiation, the OH concentration increases throughout the 1-mm area, and the maximum concentration increases by approximately 6.5 times from 0.054 to 0.36 nM. This figure indicates that UV irradiation accelerates OH generation immediately after plasma ignition. In the case of NO, the concentration is increased by more than an order of magnitude. The spatial distribution is also notably changed to a uniform distribution near the interface. This spatial distribution of the NO concentration is similar to that of the HONO concentration under UV exposure, which supports a substantial effect of HONO photolysis on NO. Unlike the two species, O_3 concentration decreases via photolysis.

Differences in the result are observed in the case of 300 s after turning on the plasma (Fig. 4b). Compared to the without UV case, the OH concentration increases from 0.23 to 0.30 nM at 0.01 mm at the liquid surface. In the region deeper than 0.5 mm, the OH concentration without UV irradiation is higher. This is because of the lower concentrations of HONO and H_2O_2 , which are reactants of ONOOH production: $\text{HONO} + \text{H}_2\text{O}_2 \rightarrow \text{ONOOH} + \text{H}_2\text{O}$. Owing to the photolysis of HONO and H_2O_2 , the production of ONOOH is decreased. Because ONOOH is a major reactant of the OH production reaction, the OH concentration without UV irradiation becomes higher in the deeper region.

In the case of NO, the NO concentration is enhanced from 0.062 to 3.7 nM at 0.01 mm and 0.015 to 1.5 nM at 1 mm, mainly via HONO photolysis. The change in O_3 concentration by photolysis is insignificant up to 0.1 mm; however, its difference between 0.1 and 3 mm is several orders of magnitude larger because of the increase in O atom concentration, which is the product of NO_3^- photolysis.

To analyze the effect of external UV irradiation quantitatively, the 2-fold and 4-fold UV intensities were additionally tested; the UV photolysis rate Q is directly proportional to the UV irradiance $F(\lambda)$ as expressed in Eq. (11). As seen in Fig. 4a, the maximum OH concentration increases

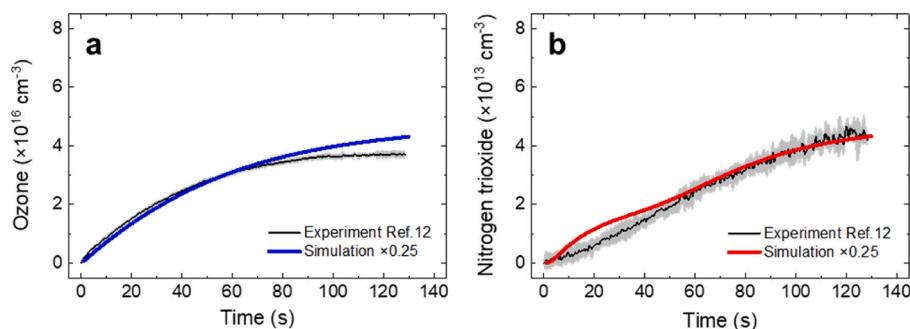


Fig. 3. Comparison of (a) O_3 and (b) NO_3 temporal concentration distributions of the measurement and simulation results at a distance of 50 mm from the SDBD plasma for model validation [29].

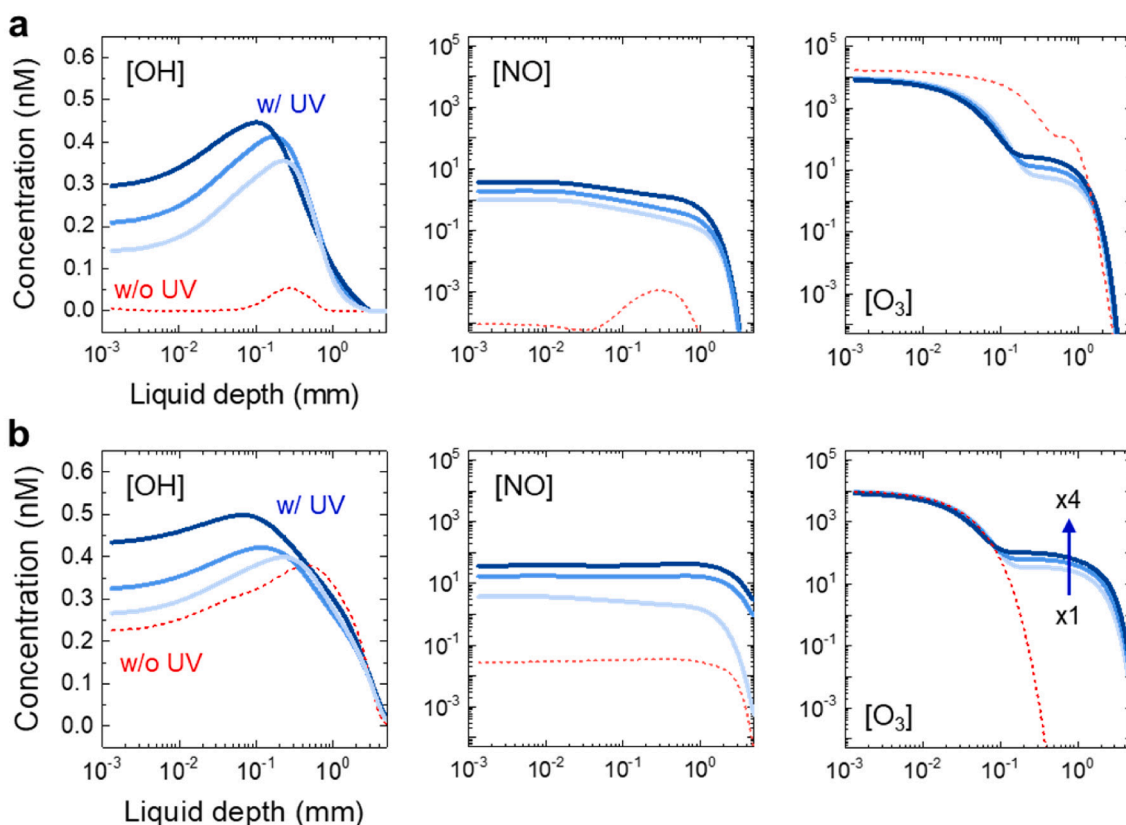


Fig. 4. Spatial concentration distributions of OH, NO, and O_3 in the liquid region at (a) 100 s and (b) 300 s after plasma ignition. The blue curves indicate the concentrations with UV irradiation, and the red dot curves represent the concentrations without UV irradiation. The RONS concentration with 1 time, 2 times, and 4 times intensity of UV irradiation is represented as color saturation to indicate the change as the UV irradiation intensity increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

up to 0.45 nM with increasing Q . Furthermore, the location of peak concentration becomes lower and the OH concentration without UV irradiation becomes higher in the region deeper than 0.5 mm.

In the case of 300 s (Fig. 4b), the OH concentration at 0.01 mm increases to 0.34 and 0.44 nM when the UV intensity is doubled and quadrupled, respectively. Compared with the case without UV exposure, the OH concentration at 0.01 mm enhances by 1.8 times under UV irradiation. The NO concentration is uniformly enhanced as UV intensity increases. At 300 s, under a quadrupled UV exposure, the NO concentration increases to 3.9 nM at 0.01 mm and 4.1 nM at 1 mm. These results demonstrate that an external UV light can be used to effectively control short-lived reactive species over the entire liquid region of PTW.

3.3. Spatiotemporal behavior of OH and NO

Now, we discuss the spatiotemporal behavior of the OH and NO in PTW. Fig. 5a shows the time evolution of the OH concentration with (blue) and without UV (red-dotted) at three distances from the liquid surface: 0.01 mm, 0.1 mm, and 1 mm. Again, the dependence of the UV intensity on OH concentration was investigated using UV intensities of 1, 2, and 4 times the measured value, as depicted in the figure.

At 0.01 mm, as shown in Fig. 5a, the OH concentrations gradually increase and reach a plateau in all cases. This result means the reactive species generated in the plasma continuously dissolve and diffuse over time near the water's surface. In the absence of UV, an increase in OH concentration was observed slowly (after 130 s). In contrast, the UV light makes the production of OH in PTW much faster. In addition, the

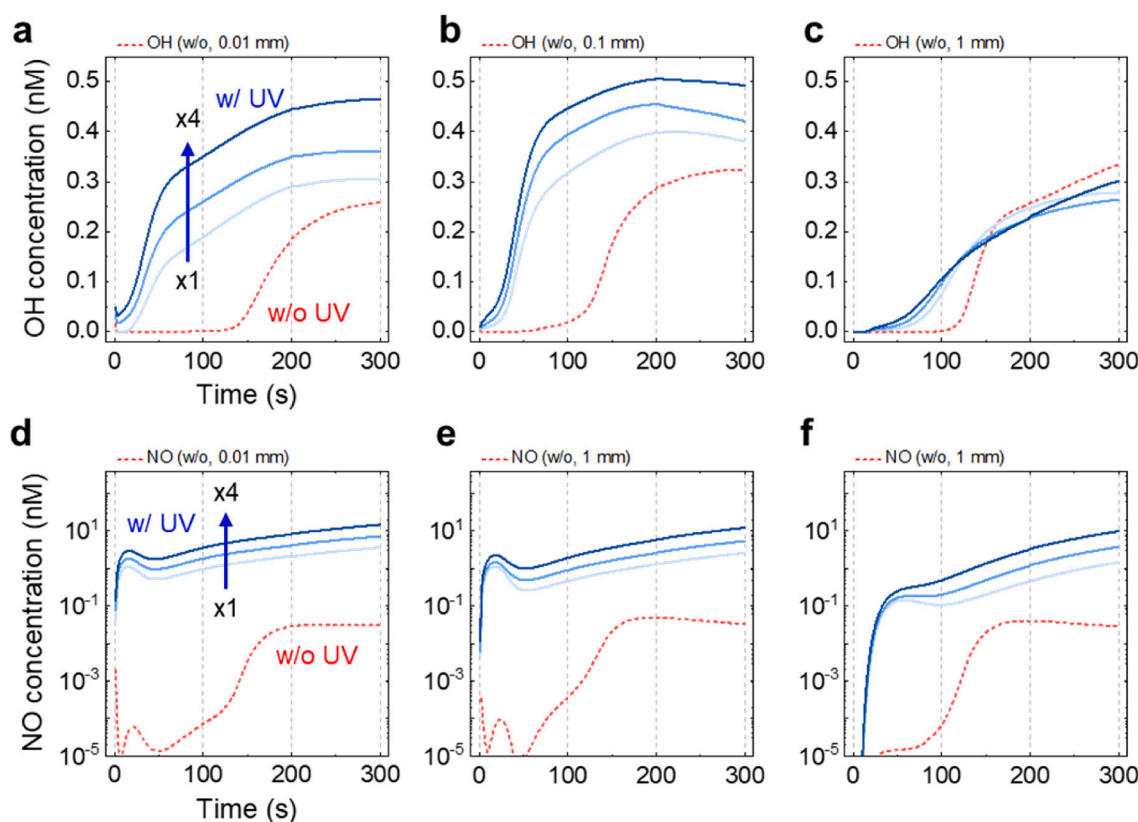


Fig. 5. Comparison of the temporal distributions of OH concentration in the liquid region at (a) 0.01 mm, (b) 0.1 mm, and (c) 1 mm. Similar to OH, the temporal distribution of NO concentration in the liquid region according to the intensity of UV irradiation and intensity are compared at (d) 0.01, (e) 0.1, and (f) 1 mm. The blue curves and red dot curves indicate the concentrations with and without UV irradiation, respectively. The RONS concentration with 1 time, 2 times, and 4 times intensity of UV irradiation is represented as color saturation to indicate the change as the UV irradiation intensity increases. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

concentration is much higher in the presence of UV throughout the entire time range—larger than 0.3 nM at 300 s. When the UV intensity is increased 2 and 4 times, the concentration increases by 1.4 and 1.8 times, respectively, at 300 s. Still, at 0.1 mm, a similar trend of OH concentrations remains, meaning that UV exposure is highly effective in improving the OH radical production in the early PTW.

At 1 mm, the temporal behavior of OH concentration differs from the previous two results. After 150 s, the concentration in the case without UV becomes higher. Owing to the increase in the concentration of HONO and H_2O_2 by dissolution, the concentration of ONOOH increases gradually over time: $\text{HONO} + \text{H}_2\text{O}_2 \rightarrow \text{ONOOH} + \text{H}_2\text{O}$. Because the decomposition of ONOOH is a major reaction of OH production at 1 mm, the difference in production rate according to the UV irradiation is reduced over time. In summary, UV irradiation shortens the time for reaching 0.1 nM of OH in the entire region, and such UV effect is significant in the region shallower than 0.5 mm but reduced in the deep region (approximately 1 mm).

To further understand the temporal behavior, the total reaction rate of OH according to the UV irradiation and intensity are compared in [Supplementary Fig. 2](#), where spatial profiles of the OH production rates at 100 and 300 s are plotted. As shown in [Supplementary Fig. 2a](#), the OH production rates with UV irradiation at 0.01 and 1 mm are 4740 and 21 nM/s at 100 s, respectively, which is 10 and 3700 times larger than those without UV irradiation (red-dotted curve). [Supplementary Fig. 2b](#) presents the enhancement of the OH production rate at 300 s. The OH production rates with UV irradiation at 0.01 and 1 mm are 10,900 and 1290 nM/s, respectively.

However, the concentration difference compared to the without UV case is not substantial because of loss reactions. Therefore, the net reaction rate of OH is approximately 2 nM/s at 0.01 mm. Based on this

result, we can compare the net OH production to our previous experiment [50]. Assuming that the volume of liquid in the experiment and simulation is the same, the estimated OH production over 600 s is approximately 30 μM comparable to the experimental result of 100 μM . These results increase the reliability of the simulation model despite different discharge conditions and the plasma-water distance.

The temporal behavior of NO changed because of UV irradiation. [Fig. 5d](#), [5e](#), and [5f](#) represent the temporal distribution of the NO concentration in the liquid region according to the UV irradiation at 0.01, 0.1, and 1 mm, respectively. As shown in [Fig. 4](#), UV irradiation enhances the NO concentration over the entire region, and the NO concentration increases in proportion to the UV intensity. In [Fig. 5d](#), under 2 and 4 times increase in UV intensity, the concentration increases by approximately 1.9 and 3.7 times, respectively. A similar temporal behavior appears at the 0.1 and 1 mm positions, as shown in [Fig. 5e](#) and [5f](#). Furthermore, dramatic differences were not observed in the NO concentration compared to the 0.01-mm case, unlike the OH concentration. UV irradiation induces an abrupt increase in the concentration in the early period. Along with this, the NO concentration increases continuously after 50 s at 0.01 and 0.1 mm and after 100 s at 1 mm under UV irradiation.

3.4. Production rate of OH

To quantitatively analyze the contribution of UV photolysis, OH production rates of each chemical reaction are compared. Note that more than 95 % of NO is formed through the HONO photolysis reaction ($\text{HONO} + h\nu \rightarrow \text{OH} + \text{NO}$) and N_2O_3 decomposition ($\text{N}_2\text{O}_3 \rightarrow \text{NO}_2 + \text{NO}$) reaction in the entire liquid region. Thus, the contribution of UV photolysis to OH production reaction will be mainly discussed.

The effect of the UV photolysis reactions on OH production in the liquid region is shown in Fig. 6. The contribution of each chemical reaction to OH production at 100 s in the liquid region at 0.01, 0.1, and 1 mm are summarized in Fig. 6a, 6b, and 6c, respectively. At 0.01 mm, the photolysis of H_2O_2 and HONO accounts for approximately 13 % and 1.8 % of the total OH production, respectively (Fig. 6a). At 0.1 mm, the UV photolysis reactions account for 37 % of the total OH production (Fig. 6b). OH production in the PTW is highly related to ONOOH, which correlates with NO and NO_2 . Similar amounts of OH are also produced by the photolysis of H_2O_2 . The HO_3 decomposition reaction is the major reaction at 0.01 mm because the HO_3 concentration increases as pH decreases. Compare to the 0.1 mm case, the contribution of UV photolysis reactions is reduced at 1 mm because of the decrease in H_2O_2 and HONO from the plasma (Fig. 6c). Accordingly, the contribution of the ONOOH decomposition reaction increases to 75 %. However, the total loss reaction rate is almost the same as the total production rate, and reactions with O_3 and NO_2 are dominant loss reactions of OH, as shown in Supplementary Fig. 3a, 3b, and 3c.

In the case of the 300 s, as shown in Fig. 6d, 6e, and 6f, the reaction rates of HONO and ONOOH become larger at both locations because of the NO enhancement by UV photolysis. Compared to the 100 s cases, the ratio of HONO photolysis to the total reaction rate increases 3.5, 4.0, and 2.8 times at 0.01, 0.1, and 1 mm, respectively. Furthermore, the contribution of the ONOOH decomposition reaction increases in the entire region. In the case of OH loss reactions, the ratio of the HONO reaction increases. Therefore, the UV-C spectrum that decomposes H_2O_2 and the UV-A spectrum that enhances NO production are crucial in the enhancement of OH production.

4. Summary and conclusion

Our first effort in this study was to make the numerical modeling of the plasma-water system more realistic by considering convective flow

and heat transfer plus water evaporation. The accuracy of our modeling result was validated by comparing it with our experimental result. Then, we successfully demonstrate how the photochemical processes induced by the external UV source notably affected the chemical kinetics in the plasma-liquid system and PTW.

In summary, UV irradiation in the liquid region enhances the total OH concentration 11.6 times and 1.1 times at 100 s and 300 s, respectively. In the entire region, H_2O_2 and HONO photolysis accounts for about 9.3 % and 2.2 % of the total OH production: $\text{H}_2\text{O}_2 + h\nu \rightarrow 2\text{OH}$, $\text{HONO} + h\nu \rightarrow \text{NO} + \text{OH}$. In the case of NO, the total concentration in the liquid region is enhanced 1,100 times and 630 times at 100 s and 300 s, respectively, and HONO photolysis accounts for about 51 % of the total production: $\text{HONO} + h\nu \rightarrow \text{OH} + \text{NO}$. Furthermore, the moment at which OH and NO concentrations increase sharply was advanced by about 100 s, and their increases are proportional to the UV irradiance.

Our findings reveal that the concentrations of OH and NO and their chemical stabilities increase under UV exposure. In light of the short lifetime of radicals in PTW, we conclude that the UV-driven gaining of OH and NO radicals in PTW is an quite efficient and simple way; lamp parameters such as exposure time and intensity can be used as independent variables to control the radical gain. In addition, wavelength-dependent photolytic processes allow selective control of the reactive species and resultant reactions required for the target to be treated in PTW.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

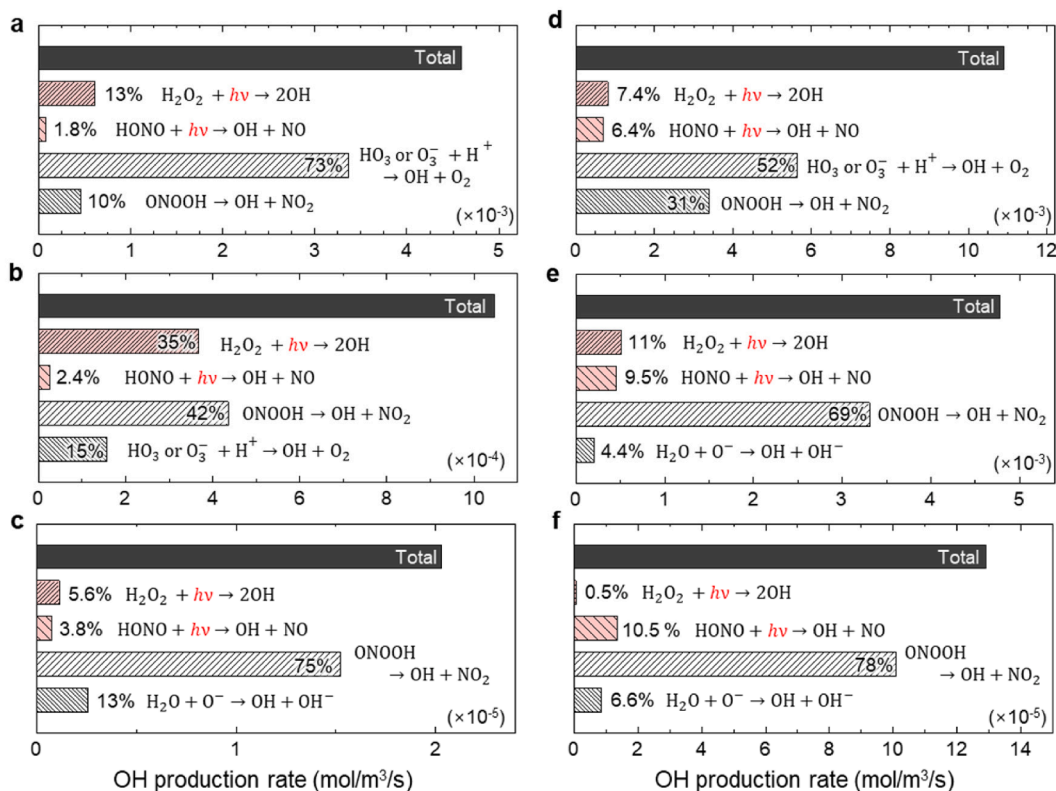


Fig. 6. The total OH production rate, individual UV photolysis reaction rates, and OH-related reaction rates in the liquid region at 100 s at (a) 0.01, (b) 0.1, and (c) 1 mm and at 300 s at (d) 0.01, (e) 0.1, and (f) 1 mm from the gas-liquid interface.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cej.2023.141425>.

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